

General Topology

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Chapter 1

Topological Spaces

Problems

A LARGEST AND SMALLEST TOPOLOGIES

- (a) The intersection of any collection of topologies for X is a topology for X .
- (b) The union of two topologies for X may not be a topology for X (unless X consists of at most two points).
- (c) For any collection of topologies for X there is a unique largest topology which is smaller than each member of the collection, and a unique smallest topology which is larger than each member of the collection.

B TOPOLOGIES FROM NEIGHBORHOOD SYSTEMS

- (a) Let $(X, \mathfrak{T})$ be a topological space and for each x in X let $\mathfrak{U}_x$ be the family of all neighborhoods of x . Then:
 - (i) If $U \in \mathfrak{U}_x$, then $x \in U$.
 - (ii) If U and V are members of $\mathfrak{U}_x$, then $U \cap V \in \mathfrak{U}_x$.
 - (iii) If $U \in \mathfrak{U}_x$ and $U \subset V$, then $V \in \mathfrak{U}_x$.
 - (iv) If $U \in \mathfrak{U}_x$, then there is a member V of $\mathfrak{U}_x$ such that $V \subset U$ and $V \in \mathfrak{U}_y$ for each y in V (that is, V is a neighborhood of each of its points).
- (b) If $\mathfrak{U}$ is a function which assigns to each x in X a non-void family $\mathfrak{U}_x$ satisfying (i), (ii), and (iii), then the family $\mathfrak{T}$ of all sets U , such that $U \in \mathfrak{U}_x$ whenever $x \in U$, is a topology for X . If (iv) is also satisfied, then $\mathfrak{U}_x$ is precisely the neighborhood system of x relative to the topology $\mathfrak{T}$.

Solutions

SOLUTION TO PROBLEM A

- (a) Let blah blah blah

